SWIM WITH TINYFISH · ROUND 1

# GhostInventory

## An AI agent that finds the supply chain's hidden inventory.

Type a part number. TinyFish agents hunt across supplier websites live, verify every spec, and return sourcing options with proof.

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THE PROBLEM

# When the usual supplier runs out, the line stops

Component shortages are not about zero stock. The stock exists, it is just hidden.

## 01

### Scattered supply

- Parts sit across authorised distributors, brokers and manufacturer stores
- Every site has a different layout
- No single clean API covers them

## 02

### Slow manual hunting

- Buyers open dozens of tabs per part
- Read datasheets to confirm package and grade
- Email suppliers for lead times

## 03

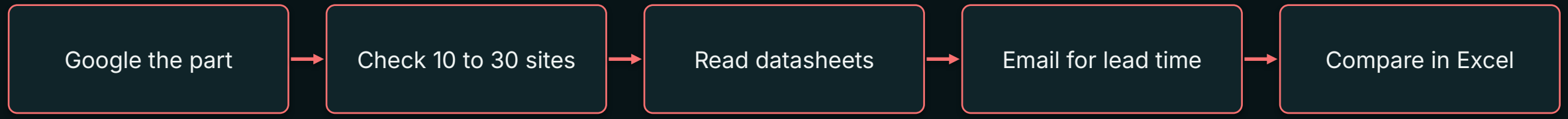
### Expensive mistakes

- Wrong package or temperature grade means rework
- Unknown brokers raise counterfeit risk
- Idle line or repair job while waiting

TODAY VS. GHOSTINVENTORY

# From days of tab-hopping to minutes

## TODAY (MANUAL)




Hours to days per hard-to-find part. Results go stale the moment they are copied.

## WITH GHOSTINVENTORY




**Goal: minutes per part, live data, and a source link behind every number.**

PRODUCT FEATURES

# What GhostInventory does

### 1 Part search by MPN or specs

Enter a manufacturer part number, or describe the part by value, package, tolerance, voltage and temperature grade.

### 2 Parallel multi-site hunt

Many TinyFish agents run at once, one per supplier site, so results arrive in minutes instead of hours.

### 3 Spec verification

Each listing is checked field by field against the required spec. Mismatches are flagged, never hidden.

### 4 Alternate finder

When the exact part is unavailable, suggests pin-compatible or parameter-equivalent replacements.

### 5 Evidence trail

Every price, stock count and lead time links to the source page, with a timestamp of when it was read.

### 6 Stock Watch

Set a watch on a sold-out part. A TinyFish monitor re-checks and alerts when stock returns.

THE SOLUTION

# Type a part number. Get verified suppliers with proof.

1

## Input

Part number or specs: value, package, tolerance, temp range, quantity

2

## Discover

TinyFish Search finds candidate suppliers beyond the big distributors

3

## Navigate

Parallel TinyFish web agents open each site, search, read stock and price tables

4

## Verify

LLM checks every listing against the exact spec and flags mismatches

5

## Deliver

Ranked shortlist, alternates, evidence links, Stock Watch

Evidence first

Every number links to the exact page the agent saw.

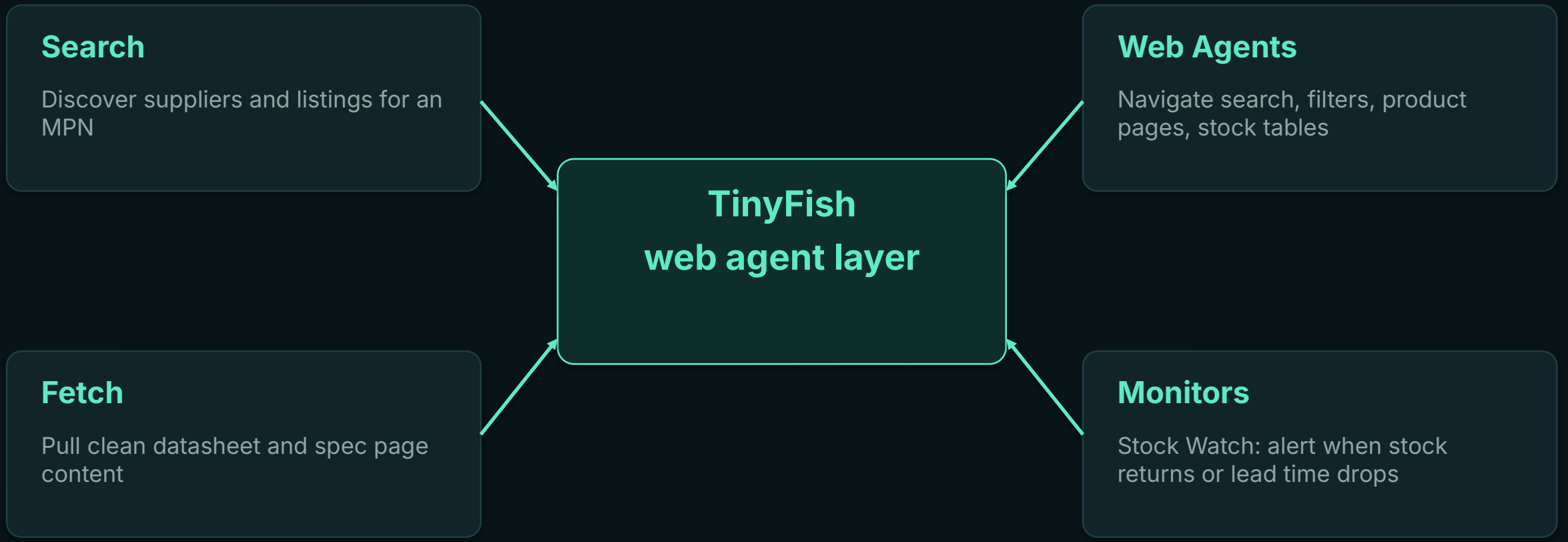
Spec-safe alternates

Suggests drop-in replacements with matching parameters.

HOW TINYFISH POWERS IT

# TinyFish is the engine, not an add-on

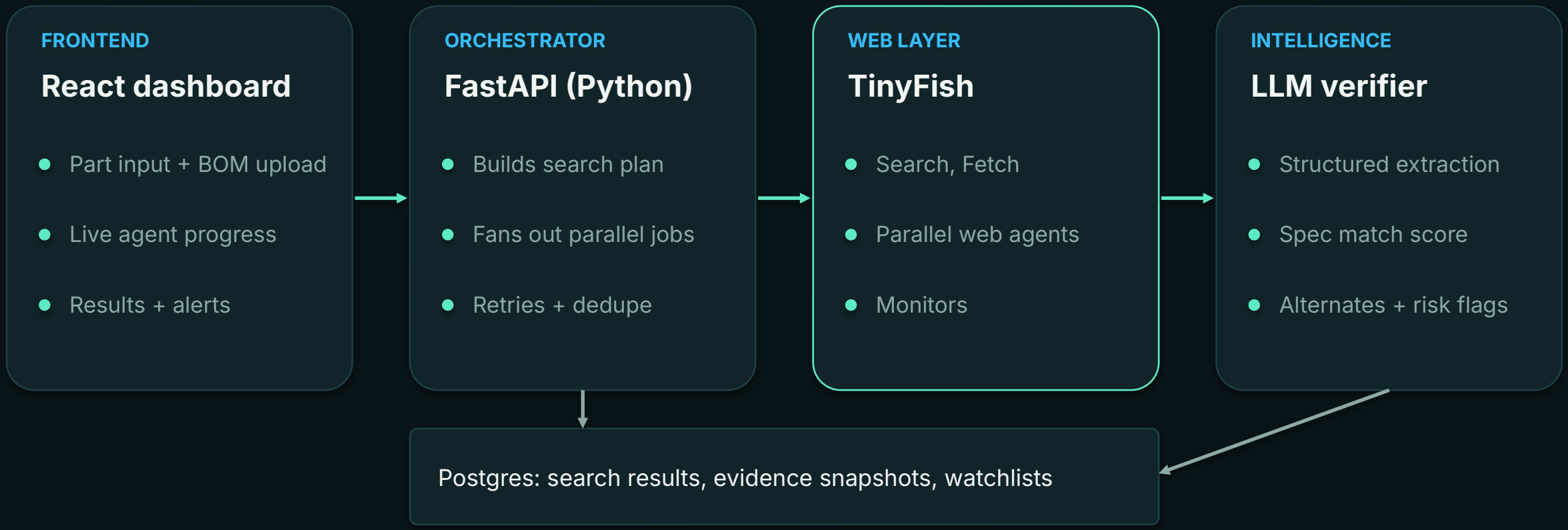
Supplier sites have no shared API, change layouts often and update stock hourly. Only live web agents can read them reliably.




**Remove TinyFish and the product does not exist.**

PLANNED ARCHITECTURE

# How the system fits together




Output per listing: supplier · MPN · stock · MOQ · price breaks · lead time · package · match score · source URL

WHAT THE USER SEES

# Sample output (illustrative mock data)

Search: STM32F103C8T6, LQFP-48, qty 500, industrial temp range

| Supplier | Type | Stock | Lead time | Unit price | Spec match | Evidence |
|---|---|---|---|---|---|---|
| Supplier A | Authorised | 0 | 26 wks | n/a | Exact | Link |
| Supplier B | Authorised | 1,200 | In stock | ₹ •• | Exact | Link |
| Supplier C | Broker | 650 | 3 days | ₹ •• | Exact, check date code | Link |
| Supplier D | Manufacturer store | 800 | In stock | ₹ •• | Alternate (pin-compatible) | Link |
| Supplier E | Broker | 5,000 | 2 days | ₹ •• | Mismatch: wrong package | Link |

Exact match  Needs a check  Rejected

Ranking = spec match + stock vs. need + lead time + supplier trust. Supplier names hidden in mock.

INSIDE ONE AGENT RUN

# What happens on each supplier site

### Open

Load supplier site in a TinyFish browser session

### Search

Type MPN, apply filters, handle pagination

### Extract

Read stock, MOQ, price breaks, lead time, package

### Normalise

Convert units, currencies and naming into one schema

### Verify

Compare against required spec, score the match

### Return

Structured JSON + source URL + timestamp

## VERIFICATION RULES

- Exact MPN match, including the suffix
- Package, pin count and temperature range must match
- Lifecycle status read from datasheet (active, NRND, EOL)
- Quantity available vs. quantity needed
- Missing field = marked unknown, never guessed

## RELIABILITY

- Per-site timeout and retry, one slow site never blocks the rest
- Duplicate listings merged across sites
- Goal-based agents survive site layout changes
- Results cached with timestamp for repeat searches
- Every step logged for a full audit trail

TEAM AND BUILD SCOPE

# Team Moorthi Annan Gang

NK

## Nidesh Kaarthik R S

TEAM LEAD

- CS undergraduate, Scaler School of Technology, Bangalore (Batch 2029)
- Focus: AI/ML, AI agents, MCP and product building
- Owns architecture, agent orchestration and build

WHAT WE WILL BUILD

### Agent core

TinyFish agents on 3 to 5 supplier sites with one shared extraction schema

### Verification engine

LLM spec matching, match score and mismatch flags

### Dashboard

Part input, live agent progress, ranked results table with evidence links

### Stock Watch

TinyFish monitor on one sold-out part with an alert